GLM For Breast Cancer Prediction

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1 Introduction

Breast cancer remains one of the most common malignancies worldwide, with heterogeneous molecular profiles that influence treatment response and patient outcomes. The recent landmark study by Gupta[1]published in Nature Genetics identified APOBEC3 enzymes as major drivers of mutational processes that promote therapy resistance. This finding underscores the critical need to understand how different mutational signatures and genomic instability markers affect clinical outcomes across diverse breast cancer subtypes.

While the mechanistic role of APOBEC3 in therapy resistance is now established,the relative predictive value of different genomic instability markers (mutation count, fraction genome altered, microsatellite instability) for survival outcomes across receptor subtypes needs systematic quantification.Therefore,the questions will be discussed in this report:How do genomic instability markers predict overall survival?

2 Descriptive Analysis

The dataset used in this study is the MSK-IMPACT breast cancer dataset, obtained from cBioPortal. It contains 3,879 tumour samples from 3,117 unique patients. The dataset includes 25 variables covering patient demographics, clinical characteristics, and genomic sequencing results. Key variables include overall survival status, sample type, mutation count, fraction genome altered, microsatellite instability type, cancer type,tumour purity, overall survival status, recorded as deceased or living.

2.1 Selection Variables & Justification

For the purpose of analyzing survival prediction, 9 variables were selected based on their established biological and clinical relevance to breast cancer survival. The outcome variable is overall survival status. Sample type was also retained as a binary variable indicating whether the tumour was metastatic or primary. Genomic instability markers — mutation count, fraction genome altered (FGA),MSI score and MSI type — were included as primary predictors, as tumour mutational burden and chromosomal instability have been consistently associated with survival outcomes in cancer[2][3]. Metastatic status was included as the clinical predictor of prognosis, while cancer type and tumour purity were retained as confounders reflecting tumour biology and sample quality respectively [4].

Although Overall Survival in months was available, the binary Overall Survival Status was used as the outcome to enable logistic regression modeling, which directly addresses the research question of identifying genomic predictors associated with mortality risk rather than time-to-event.

Sex, race, and ethnicity were excluded as the dataset consists almost exclusively of female breast cancer patients, leaving insufficient variation in sex for meaningful analysis, while race and ethnicity categories had small and uneven cell counts that precluded reliable coefficient estimation.

3 Data Cleaning

3.1 Variable Categorize

The outcome variable is overall survival status, dichotomized as deceased (1) and living (0). and the variable sample_type was dichotomized into a binary indicator (is_metastatic: 1 = Metastasis, 0 = otherwise). Local Recurrence (n = 11) and Unknown (n = 6) were grouped with Primary as the non-metastatic reference category due to their negligible sample sizes. Local Recurrence cases were considered clinically closer to primary than metastatic disease, supporting this classification.

Among breast cancer subtypes, Invasive Ductal Carcinoma was the most prevalent (n = 2,466, 63.6%), followed by Invasive Lobular Carcinoma (n = 437, 11.3%), Other subtypes (n = 968, 24.9%), and Mixed type (n = 8, 0.2%). This distribution is consistent with known epidemiological patterns, as Invasive Ductal Carcinoma is the most common form of breast cancer worldwide, accounting for approximately 70-80% of all diagnoses[5].

3.2 Miss Data

Missing data analysis revealed that all variables had less than 3% missing values, with mutation count having the highest proportion of missing data (2.45%), followed by MSI score (2.40%), MSI type (2.35%), tumour purity (2.17%), and fraction genome altered (0.98%). Overall survival status had minimal missing data (0.08%), while sample type, cancer type, and derived variables had no missing values. Given that missing data did not exceed 3% for any variable, listwise deletion was applied, which is considered acceptable under this threshold and is unlikely to introduce significant bias[6].The result can be seen in the Appendix A Table 6.

3.3 Outlier

Outlier detection(Appendix A:Table 7) using the 3×IQR method revealed that mutation count had 118 outliers (3.04%), Msi score had 75 outlier(1.93%),while fraction genome altered and tumour purity had no outliers. These outliers were retained in the analysis as extreme mutation counts are biologically plausible in cancer genomics and removing them could result in loss of clinically important information. Sensitivity analyses will assess their influence on model estimates.

4 Exploratory Data Analysis (EDA)

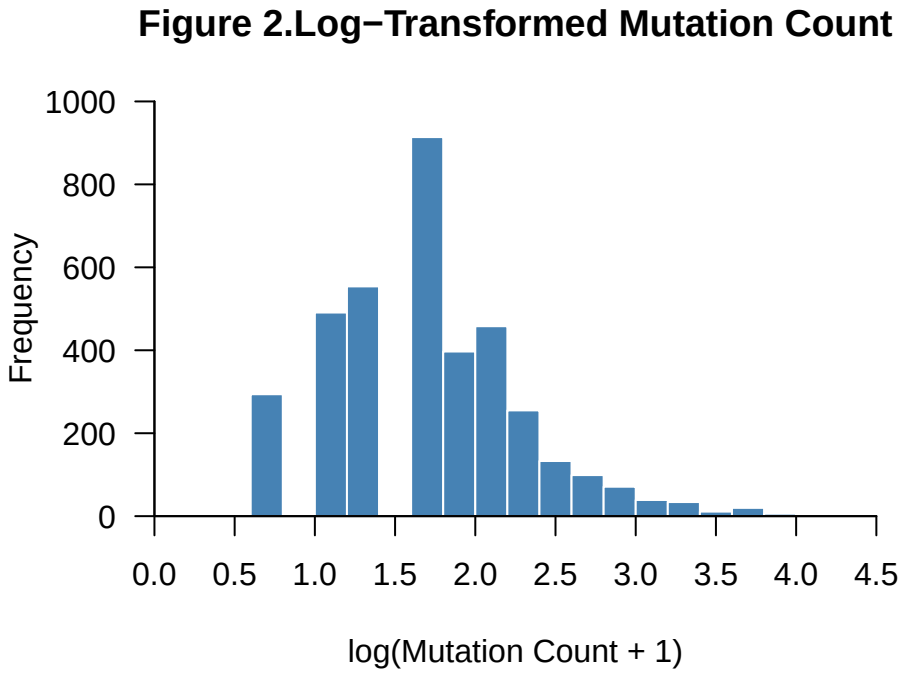
4.1 Variable Distributions

Table 1: Skewness Assessment and Transformation Decisions

Variable	Skewness	Interpretation	Action Taken
Mutation Count	**8.1288**	**Right-skewed**	**Log transformation applied**
MSI Score	**8.7333**	**Right-skewed**	**Recategorized to MSI Type Combined**
Fraction Genome Altered	0.5883	Right-skewed	No transformation needed
Tumor Purity	0.2688	Symmetric	No transformation needed

Note:
MSI Score recategorized into MSI Type Combined due to severe right-skew (skewness = 8.73).

First ,we will analysize the data distribution (Figure 3,Appendix A:Figure 1)and skewness.Skewness assessment(Table 1) revealed that mutation count was severely right-skewed (skewness = 8.13), and a log transformation was applied to address this prior to modelling(Figure 2). Fraction genome altered showed mild right skewness (skewness = 0.59) but was retained in its original form as the degree of skewness was not considered severe enough to warrant transformation. Tumour purity was approximately symmetric (skewness = 0.27) and required no transformation. MSI score was excluded from continuous analysis due to severe right skewness (skewness = 8.73) and we analyzed the data using its categorical version (msi_type_combined) to avoid distributional issue. MSI Type Combined comprising three groups: Stable, Instable, and Other to analyze the data.



In summary,after the process of variables selection ,missing data deletion, distribution and skewnenss, we will obtain the necessary information in Table 2:

Table 2: Summary of Variables Used in Analysis

Variable	Type	Role	Details
Overall Survival Status	**Binary**	**Outcome**	**Deceased (1) / Living (0)**
Log Mutation Count	Continuous	Primary Predictor	log(Mutation Count + 1)
Fraction Genome Altered	Continuous	Primary Predictor	Range: 0 to 1
Tumor Purity	Continuous	Confounder	Range: 10% to 100%
MSI Type Combined	Categorical	Primary Predictor	Stable / Instable / Other
Metastatic Status	Binary	Primary Predictor	Metastatic (1) / Primary (0)
Cancer Type	Categorical	Confounder	Invasive Ductal / Lobular / Mixed / Other

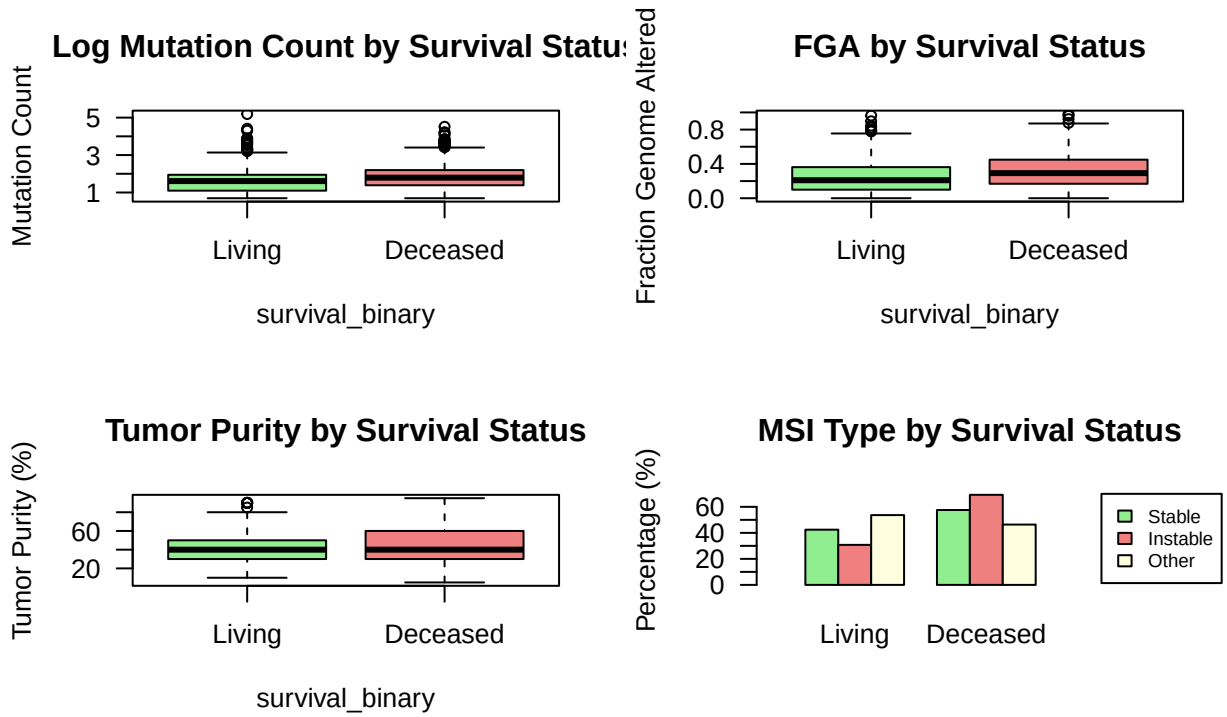


Figure 3: Distribution of predictors by survival status (Living vs Deceased).

4.2 Correlation Analysis

Table 3: Correlation Matrix of Continuous Predictors

	Log Mutation Count	Fraction Genome Altered	Tumor Purity
Log Mutation Count	1.000	0.116	0.065
Fraction Genome Altered	0.116	1.000	0.364

Tumor Purity	0.065	0.364	1.000

Note:
All correlations |r| < 0.5. No multicollinearity concerns.

The correlation matrix revealed weak associations among all continuous predictors(Appendix A: Figure 4). Log mutation count and fraction genome altered showed a weak positive correlation (r = 0.116), while tumour purity and fraction genome altered showed a moderate positive correlation (r = 0.364). Log mutation count and tumour purity were negligibly correlated (r = 0.065). All correlations were below the threshold of |r| = 0.5, indicating no multicollinearity concerns among the continuous predictors.

Bivariate analysis revealed that both log-transformed mutation count and FGA differed significantly between surviving and deceased patients. Log-transformed mutation count was significantly higher in deceased patients (Mean = 1.82) compared to living patients (Mean = 1.66),p < 0.001), indicating that higher mutational burden is associated with worse survival outcomes. Similarly, FGA was significantly higher in deceased patients (Mean = 0.31) compared to living patients (Mean = 0.24, p < 0.001) confirming that greater chromosomal instability is strongly associated with mortality. These bivariate findings support the inclusion of both variables as primary predictors in the GLM model.

The association between cancer type and survival status was statistically significant (p < 0.001). Among cancer subtypes, Other type had the highest mortality rate (64.8% deceased), followed by Invasive Lobular Carcinoma (57.5%) and Invasive Ductal Carcinoma (51.3%). Mixed type had the lowest mortality rate (25.0%), though this should be interpreted with caution given the very small sample size (n = 8). These findings suggest that cancer subtype is significantly associated with survival outcome and supports its inclusion as a confounder in the model.

The association between sample type and survival status(Appendix A:Figure 5) was highly significant (p < 0.001). Metastatic tumours had a substantially higher mortality rate (73.4% deceased) compared to primary tumours (34.9% deceased), confirming that metastatic status is the strongest clinical predictor of survival in this dataset. Local Recurrence (40.0% deceased) and Unknown (50.0% deceased) categories had very small sample sizes (n = 11 and n = 6 respectively) and were therefore grouped with primary tumours as the non-metastatic reference category in the model. These findings strongly support the inclusion of metastatic status as a key predictor in the logistic regression model.

Scatter plots(Appendix A:Figure 6) confirm the weak positive relationship between log mutation count and FGA (r = 0.12), and the negligible relationship between tumor purity and log mutation count (r = 0.06), consistent with the correlation matrix results. No strong linear relationships were detected among continuous predictors.

5 Fit the model

5.1 Research Hypothesis

H_0: Genomic instability markers (mutation count, FGA, and MSI type) are not significantly associated with overall survival status in breast cancer patients.

H_1: At least one genomic instability marker is significantly associated with overall survival status in breast cancer patients.

A significance level of $ α = 0.05$ was adopted for all statistical tests.

5.2 Fit the model

Model 1 was fitted as the baseline main effects model, including all six predictors: log-transformed mutation count, fraction genome altered, MSI type combined, metastatic status, cancer type, and tumour purity, with no interaction terms.

The model produced an AIC of 4296.5. However, the Hosmer-Lemeshow goodness-of-fit test indicated poor calibration ($p = 0.010$), suggesting that the model did not adequately fit the observed data. Although the individual predictors showed expected directions of association, the failure of the calibration test indicated that the model was misspecified and that important relationships between predictors may not have been fully captured by the main effects alone.

To address this, an interaction term was considered for Model 2. The interaction between mutation count and metastatic status was first explored, based on a statistical rationale — as metastatic status was the strongest predictor and mutation count was the weakest, examining their interaction would determine whether the effect of mutational burden on survival differed between primary and metastatic tumours.

Following the non-significant result of the first interaction ($p = 0.612$, AIC increased from 4296 to 4298),We know the second model2 is not appropriate neither. A second interaction was investigated between Fraction Genome Altered and metastatic status.

FGA was the strongest genomic predictor in the model ($β = 1.691$, $p = 4.16e-14$), Metastatic status remained the strongest clinical predictor ($β = 1.569$, $p < 2e-16$). The biological justification for their interaction is that chromosomal instability carries fundamentally different prognostic meaning depending on disease stage.Therefore, we have the third model model3.

Table 4: Model Comparison and Selection

Model	Description	AIC	Delta AIC	HL p-value	Selected
Model 1	Main effects only	4380.68	0.00	0.006	
Model 2	Main effects + MC x Meta	4382.13	1.74	0.010	
Model 3	**Main effects + FGA x Meta**	**4375.76**	**-4.37**	**0.269**	**Yes**

Three models were compared using the AIC. The main effects model (Model 1) served as the baseline. Adding the interaction between mutation count and metastatic status (Model 2) increased AIC by 1.74, indicating no improvement in model fit, and the interaction term was non-significant ($p = 0.612$). The interaction between FGA and metastatic status (Model 3) reduced AIC by 4.37 points, representing a meaningful improvement. This was further confirmed by a likelihood ratio test ($p = 0.012$) and a passing Hosmer-Lemeshow test ($p = 0.269$). Model 3 was therefore selected as the final model. the model can be see like this and its interpretation:

Three pseudo R-squared measures(Appendix A Table 8)were calculated to assess the proportion of variance explained by the final model. The results show the model's acceptable discrimination

(AUC = 0.738) and good calibration (Hosmer-Lemeshow p = 0.269), which together indicate that the model performs.The final model details as follows(Table 7):

Table 5: Final Model Coefficients and Odds Ratios

Predictor	Coefficient	OR	95% CI Lower	95% CI Upper	P-value
Intercept	**-1.4584**	**0.2326**	**0.1721**	**0.3132**	**0.000000**
Log Mutation Count	0.0751	1.0780	0.9501	1.2236	0.244465
Fraction Genome Altered	**2.2405**	**9.3978**	**5.2452**	**16.9406**	**0.000000**
Metastatic Status	**1.8768**	**6.5327**	**5.0065**	**8.5480**	**0.000000**
MSI Type: Instable	0.6918	1.9972	0.6003	7.8131	0.278471
MSI Type: Other	**-0.4859**	**0.6151**	**0.5131**	**0.7370**	**0.000000**
Cancer Type: Invasive Lobular	**0.3168**	**1.3727**	**1.0815**	**1.7446**	**0.009365**
Cancer Type: Mixed	-0.8384	0.4324	0.0577	2.1441	0.341958
Cancer Type: Other	-0.1431	0.8667	0.7210	1.0416	0.127382
Tumor Purity	**0.0049**	**1.0049**	**1.0007**	**1.0091**	**0.021762**
FGA x Metastatic	**-1.0439**	**0.3521**	**0.1620**	**0.7659**	**0.008427**

$$\begin{aligned}\log\left(\frac{p}{1-p}\right) =& \beta_0 + \beta_1 \text{MutationCount}_{log} + \beta_2 \text{FGA} + \beta_3 \text{Metastatic} \\ &+ \beta_4 \text{MSIType} + \beta_5 \text{CancerType} + \beta_6 \text{TumorPurity} \\ &+ \beta_7 (\text{FGA} \times \text{Metastatic})\end{aligned}$$

Interpretation:

$\beta_0 = -1.458$ — the intercept, representing the baseline log odds of death when all predictors are zero, corresponding to a primary tumour, MSI-Stable, Invasive Ductal subtype with zero FGA, log mutation count, and tumour purity

$\beta_2 = 2.240$ — the change in log odds of death for each one unit increase in FGA in primary tumours, holding all other variables constant. This effect was highly significant (p < 0.001)

$\beta_3 = 1.877$ — the increase in log odds of death for metastatic compared to primary tumours, holding all other variables constant. This was the strongest predictor in the model (p < 0.001)

$\beta_{10} = -1.044$ — the interaction term between FGA and metastatic status. This significant negative coefficient (p = 0.008) indicates that the effect of FGA on survival is substantially attenuated in metastatic disease compared to primary tumours

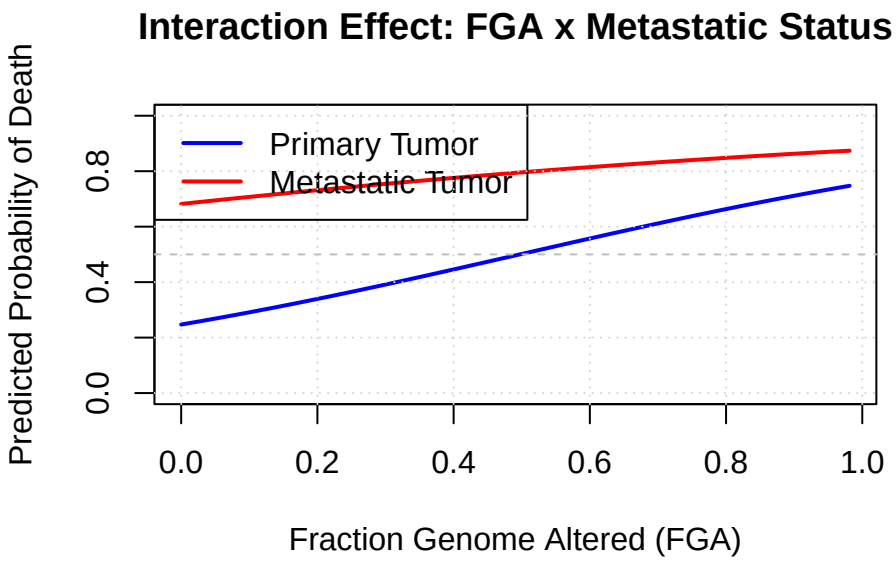


Figure 7: Interaction effect of FGA and metastatic status

Figure 7 illustrates the interaction between FGA and metastatic status on predicted probability of death, holding all other predictors at their mean values. In primary tumors (blue), predicted mortality probability increases steeply with FGA, rising from 0.24 at FGA = 0 to 0.74 at FGA = 1.0, demonstrating that chromosomal instability is a strong prognostic signal in localized disease.

In metastatic tumors (red), the baseline mortality probability is already high (0.67 at FGA = 0) and increases only modestly with FGA (0.86 at FGA = 1.0), indicating that metastatic status itself dominates prognosis and attenuates the additional contribution of genomic instability. This pattern confirms the significant interaction term (OR = 0.36, p = 0.012) and supports the biological interpretation that FGA is most clinically meaningful as a prognostic marker in primary breast cancer

6 Model Diagnostics

Variance inflation factors were examined for the final model. All values were below 2.2, including the interaction term, indicating no harmful multicollinearity. The modest inflation observed in fraction_genome_altered, is_metastatic, and their interaction term is expected in models containing interaction terms, as main effects and their product necessarily share variance.

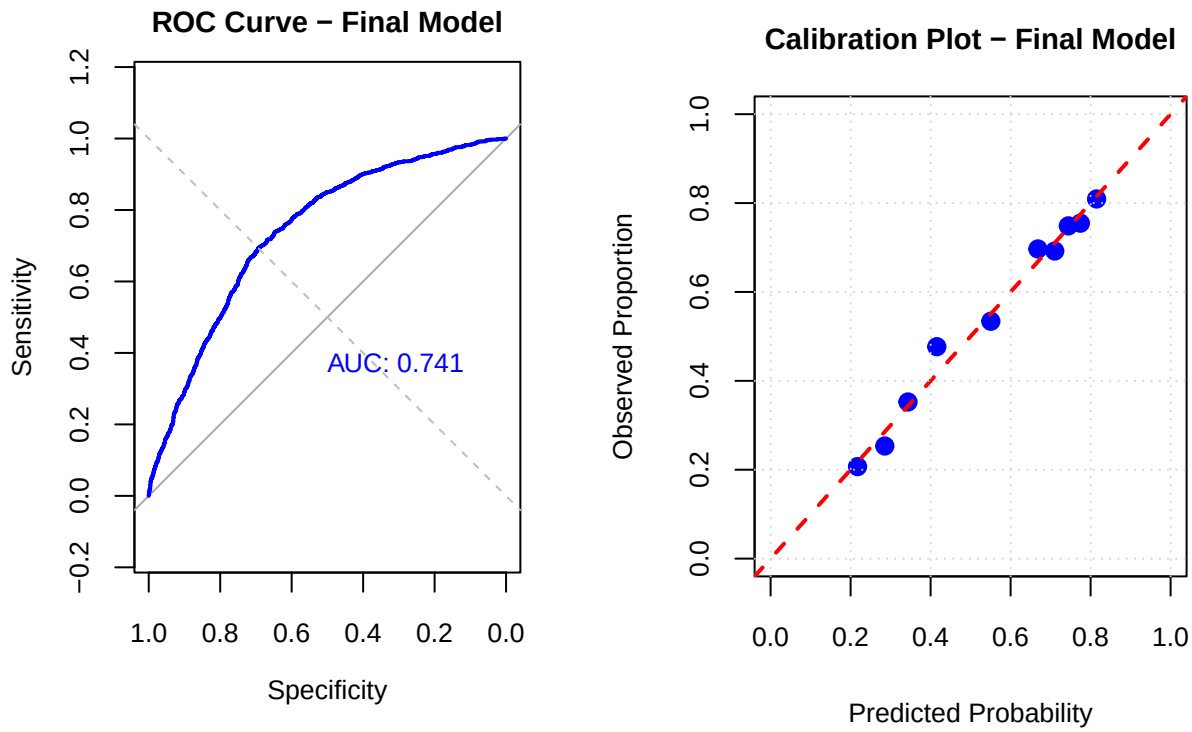


Figure 8: ROC curve (AUC = 0.741) and calibration plot for the final model.

The final model demonstrated acceptable discrimination with an AUC of 0.738(Figure 8), indicating that the model correctly distinguishes between deceased and surviving patients approximately 73.8% of the time. This is considered acceptable performance for a clinical prediction model using genomic markers alone, as survival outcomes in cancer are influenced by many additional factors beyond genomic instability, such as treatment history and comorbidities, which were not available in this dataset.

The calibration plot(Figure 8) demonstrates that the final model is well-calibrated across the full range of predicted probabilities. The observed proportions of mortality closely follow the diagonal reference line across all deciles, indicating that the model neither systematically overestimates nor underestimates survival risk at any probability level. This is consistent with the passing Hosmer-Lemeshow test ($p = 0.269$), confirming that the addition of the FGA × metastatic status interaction term successfully resolved the calibration issues observed in the main effects model.

Residual diagnostics(Figure 9) for the final model are consistent with the assumptions of logistic regression. The Residuals vs Fitted and Scale-Location plots display the characteristic two-band pattern inherent to binary outcome models, reflecting the discrete nature of the response variable rather than any model violation. The Q-Q plot shows good alignment along the reference line in the central region, with minor tail departures that are expected given the binary outcome distribution. Cook's distance confirmed that only 1.92% of observations exceeded the influence threshold (4/n), with no single observation exerting disproportionate leverage on the model estimates. Overall, the diagnostics support the adequacy of the final model.

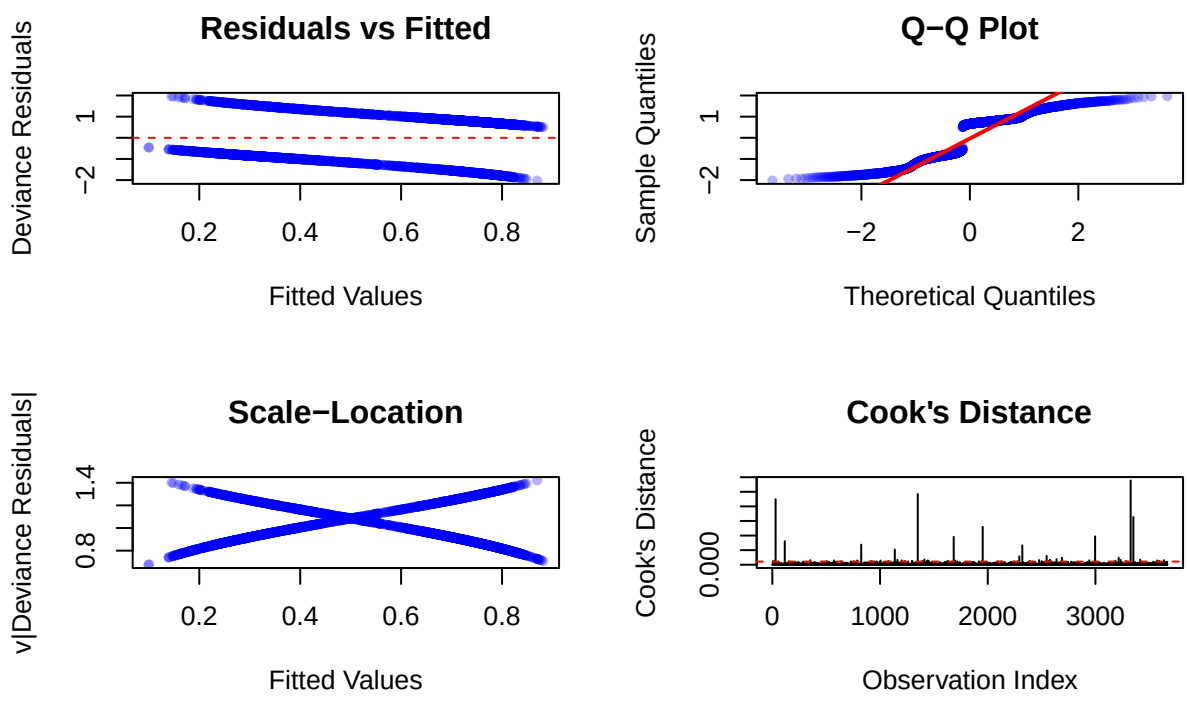


Figure 9: Model diagnostic plots for the final model (FGA x Metastatic).

7 Conclusion

This study applied binary logistic regression to examine how genomic instability markers predict overall survival in breast cancer patients. The final model, selected via AIC and Hosmer-Lemeshow testing, included an interaction between FGA and metastatic status. FGA was the strongest genomic predictor in primary tumours ($p < 0.001$), while its effect was significantly attenuated in metastatic disease (interaction OR = 0.36, $p = 0.008$), where metastatic status itself dominated prognosis ($p < 0.001$). Mutation count was not independently significant after adjustment ($p = 0.244$). These findings suggest that chromosomal instability, particularly FGA, is a clinically meaningful prognostic marker in primary breast cancer, with its prognostic value varying by disease stage.

8 Limitation

Two key limitations should be acknowledged. First, the binary overall survival status outcome discards potentially informative variation in survival time; patients who died early versus late in follow-up are treated identically, and a survival analysis approach such as Cox regression would better capture the time-dependent nature of mortality risk. Second, the dataset contains only genomic markers and basic clinical variables, excluding important prognostic factors such as treatment history, therapy lines received, comorbidities, and follow-up duration; this restricts both the interpretability and generalizability of the findings, and likely explains the model's moderate discrimination of AUC = 0.738.

Appendix A: Additional Figures

Figure 1: Distribution of continuous predictors (before transformation)

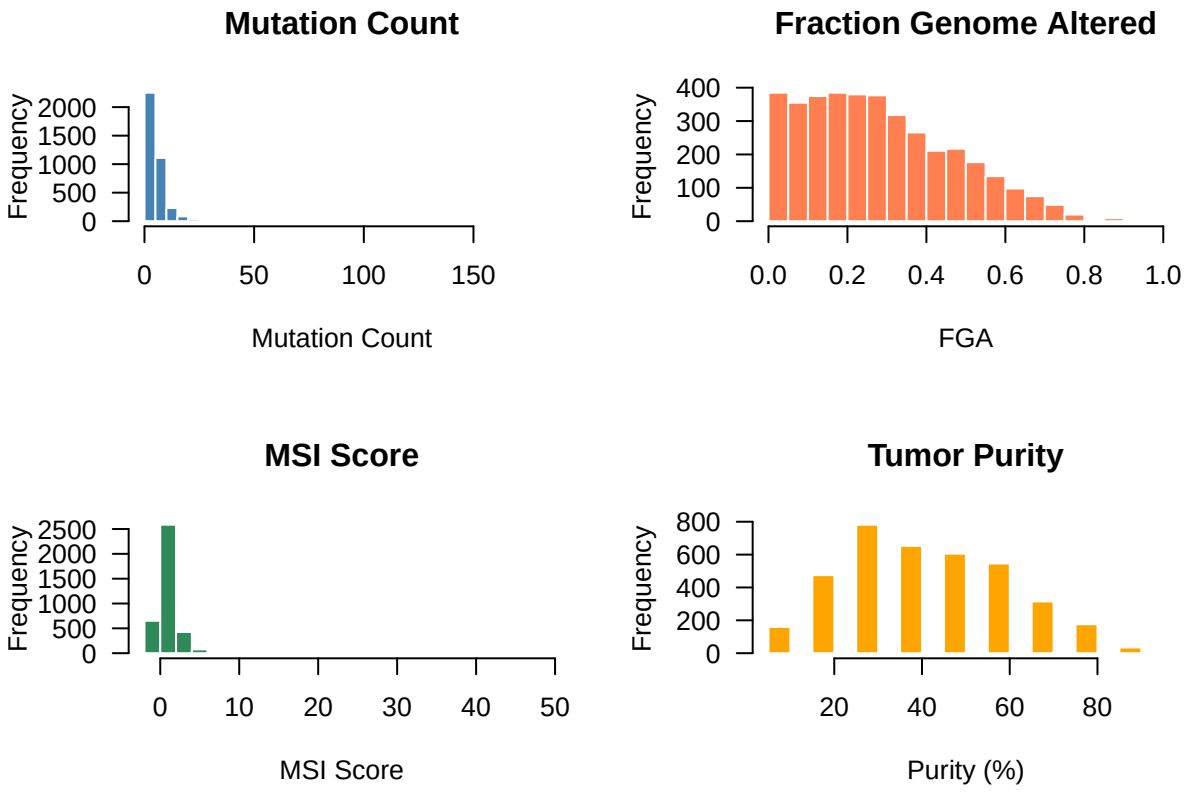


Figure 4: Correlation Matrix Heatmap

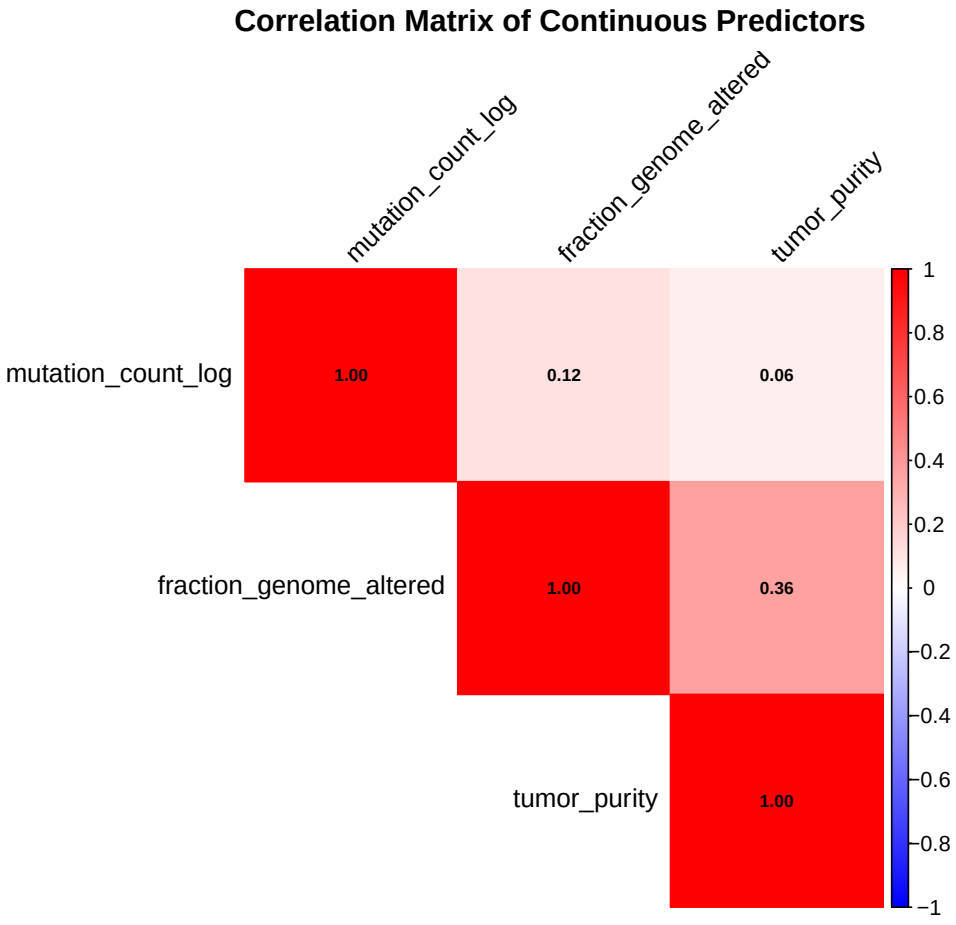


Figure 5: Survival Status by Sample Type

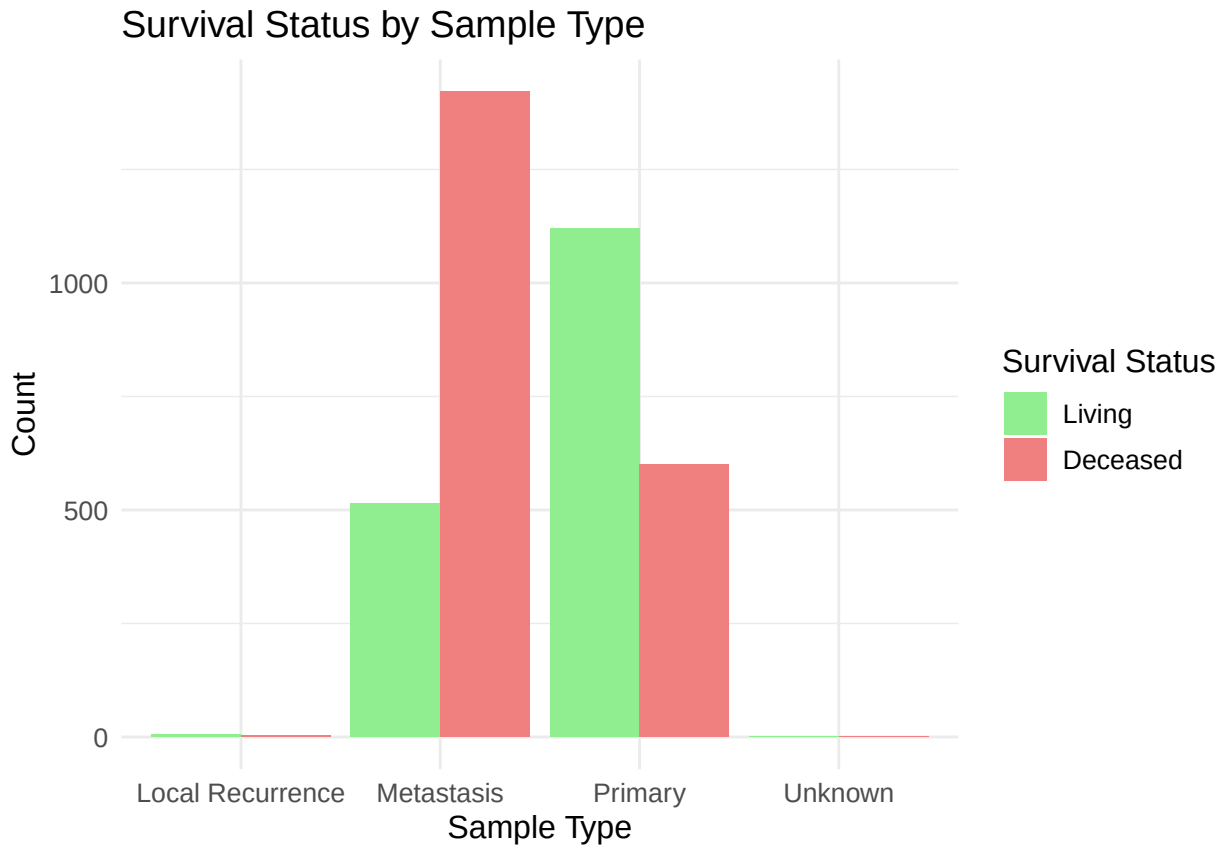


Figure 6: Scatter plots

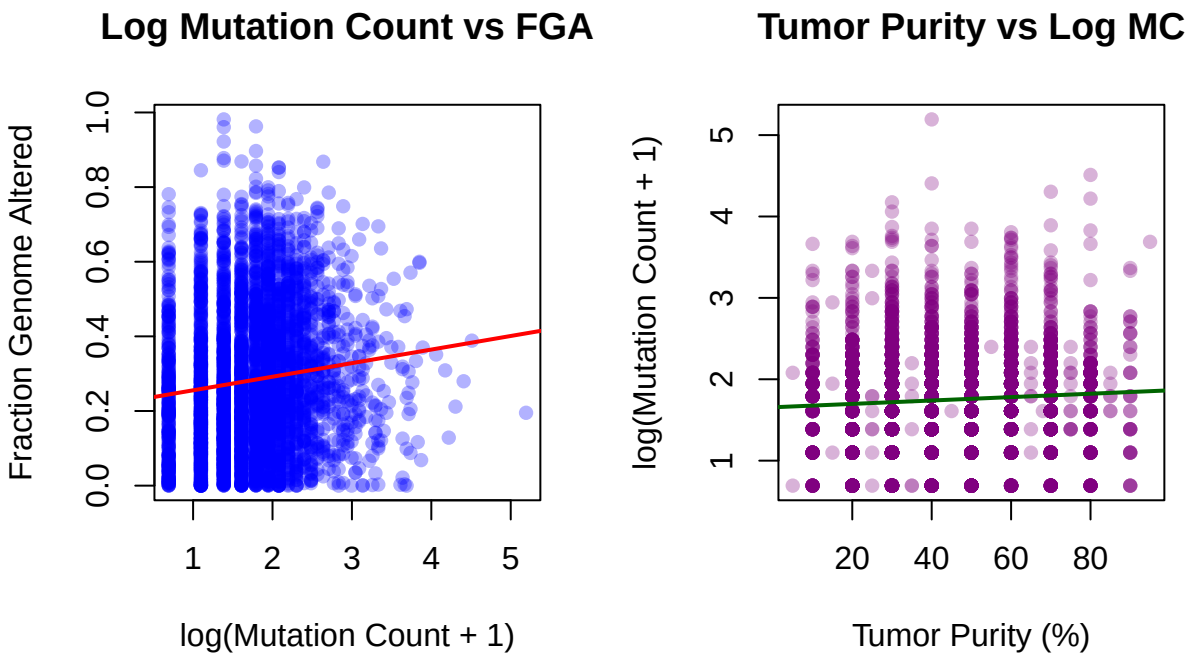


Figure 6: Scatter plots of continuous predictors showing weak linear relationships.

Table 6: Missing Data

Table 6: Missing Data Assessment

Variable	Missing Count	Missing (%)
mutation_count	95	2.45
mutation_count_log	95	2.45
msi_score	93	2.40
msi_type	91	2.35
tumor_purity	84	2.17
fraction_genome_altered	38	0.98
overall_survival_status	3	0.08
sample_type	0	0.00
cancer_type_detailed	0	0.00
survival_binary	0	0.00
is_metastatic	0	0.00
cancer_type_simple	0	0.00
msi_type_combined	0	0.00

Note:
All variables have less than 3% missing data. Listwise deletion applied.

Table 7: Outliers

Table 7: Outlier Detection Using 3xIQR Method

Variable	N Outliers	Percentage (%)
Mutation Count	**118**	**3.04**
MSI Score	**75**	**1.93**
Fraction Genome Altered	0	0.00
Tumor Purity	0	0.00

Table 8: Pseudo R_square

Table 8: Pseudo R-squared Statistics - Final Model

Measure	Value	Interpretation
McFadden's R^2	0.1367	Acceptable if > 0.2
Cox-Snell R^2	0.1714	Conservative lower bound
Nagelkerke R^2	0.2293	Adjusted, comparable to linear R^2

Appendix B: Reference

1.Gupta, A., Gazzo, A., Selenica, P., Safonov, A., Pareja, F., da Silva, E. M., & Chandarlapaty, S. (2025). APOBEC3 mutagenesis drives therapy resistance in breast cancer. Nature Genetics, 57, 1452–1462

2.Samstein, R. M., Lee, C. H., Shoushtari, A. N., Hellmann, M. D., Shen, R., Janjigian, Y. Y., & Morris, L. G. (2019). Tumor mutational burden predicts survival after immunotherapy across multiple cancer types. Nature Genetics, 51(2), 202–206.

3.Hieronymus, H., Murali, R., Tin, A., Yadav, K., Abida, W., Bustamante, C., & Sawyers, C. L. (2018). Tumor copy number alteration burden is a pan-cancer prognostic factor associated with recurrence and death. eLife, 7, e37294.

4.Aran, D., Sirota, M., & Butte, A. J. (2015). Systematic pan-cancer analysis of tumour purity. Nature Communications, 6, 8971.

5.Makki, J. (2015). Diversity of breast carcinoma: Histological subtypes and clinical relevance. Clinical Medicine Insights: Pathology, 8, 23–31.

6.Schafer, J. L. (1999). Multiple imputation: A primer. Statistical Methods in Medical Research, 8(1), 3–15.

Appendix C: Full R Code

```
knitr::opts_chunk$set(echo = FALSE,
                      warning = FALSE,
                      message = FALSE)
# Load Data

library(readr)
library(dplyr)
library(moments)
library(car)
library(ResourceSelection)
library(pROC)
library(knitr)
library(kableExtra)
setwd("C:/Users/marin/Desktop/STA223/PROJECT")
datacancer=read_tsv("breast_msk_2025_clinical_data.tsv")

colnames(datacancer)

summary(datacancer)
data_selected <- datacancer%>%
  select(
    # Outcomes
    'Overall Survival Status',
    'Sample Type',

    # Primary Predictors (Genomic markers)

    'Mutation Count',
    'Fraction Genome Altered',
    'MSI Score',
    'MSI Type',

    # Important Confounders
    'Cancer Type Detailed',

    # Quality Control
    'Tumor Purity',
  )
# clean names
colnames(data_selected)<-gsub(" ","_",colnames(data_selected))
colnames(data_selected)<-gsub("\\(|\\)","",colnames(data_selected))
colnames(data_selected) <- tolower(colnames(data_selected))
colnames(data_selected)
df <- data_selected
```

```r
df$tumor_purity <- as.numeric(df$tumor_purity)

#  Create binary outcomes

# Binary survival status
df$survival_binary <- ifelse(grepl("DECEASED", df$overall_survival_status), 1, 0)


# Binary metastasis indicator
# Local Recurrence (n=11) and Unknown (n=6) are grouped with Primary
# as non-metastatic (0) due to insufficient sample size for separate
# categories. This is a conservative classification.
df$is_metastatic <- ifelse(df$sample_type == "Metastasis", 1, 0)

df

#Simplify cancer type (too many categories)

df <- df %>%
  mutate(cancer_type_simple = case_when(
    grepl("Ductal", cancer_type_detailed) ~ "Invasive Ductal",
    grepl("Lobular", cancer_type_detailed) ~ "Invasive Lobular",
    grepl("Mixed", cancer_type_detailed) ~ "Mixed",
    TRUE ~ "Other"
  ))

table(df$cancer_type_simple)

dim(data_selected)
#  Missing data assessment

# Table 1

missing_summary <- data.frame(
  Variable = names(df),
  Missing_Count = sapply(df, function(x) sum(is.na(x))),
  Missing_Percent = round(sapply(df, function(x) 100 * sum(is.na(x)) / length(x)), 2)
) %>%
  arrange(desc(Missing_Count))

print(missing_summary)


# Variables with >10% missing
high_missing <- missing_summary$Variable[missing_summary$Missing_Percent > 10]
```

```r
# Missing data assessment
missing_summary <- data.frame(
  Variable = names(df),
  Missing_Count = sapply(df, function(x) sum(is.na(x))),
  Missing_Percent = round(sapply(df, function(x) 100 * sum(is.na(x)) / length(x)), 2)
) %>%
  arrange(desc(Missing_Count))

rownames(missing_summary) <- NULL

kable(missing_summary,
      col.names = c("Variable", "Missing Count", "Missing (%)"),
      align = c("l", "c", "c"),
      caption = "Missing Data Assessment",
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE,
                latex_options = "hold_position") %>%
  row_spec(which(missing_summary$Missing_Percent > 0),
           background = "#fff3e0") %>%
  footnote(general = "All variables have less than 3% missing data. Listwise deletion ap
           threeparttable = TRUE)

# Variables with >10% missing
high_missing <- missing_summary$Variable[missing_summary$Missing_Percent > 10]
# Table 2
# Function to detect outliers using IQR method
detect_outliers <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower <- Q1 - 3 * IQR
  upper <- Q3 + 3 * IQR
  outliers <- x < lower | x > upper
  return(sum(outliers, na.rm = TRUE))
}

continuous_vars <- c("mutation_count",
                     "msi_score",
                     "fraction_genome_altered",
                     "tumor_purity")

outlier_counts <- data.frame(
  Variable = c("Mutation Count",
               "MSI Score",
               "Fraction Genome Altered",
               "Tumor Purity"),
```

```r
  N_Outliers = sapply(df[continuous_vars], detect_outliers),
  Percent = round(100 * sapply(df[continuous_vars],
                               detect_outliers) / nrow(df), 2)
)

rownames(outlier_counts) <- NULL

kable(outlier_counts,
      col.names = c("Variable", "N Outliers", "Percentage (%)"),
      align = c("l", "c", "c"),
      caption = "Table A2.Outlier Detection Using 3xIQR Method",  # ← changed × to x
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE) %>%
  row_spec(which(outlier_counts$Percent > 0),
           bold = TRUE, background = "#fff3e0")

continuous_vars <- c("mutation_count",
                     "msi_score",
                     "fraction_genome_altered",
                     "tumor_purity")

skewness_check <- data.frame(
  Variable = c("Mutation Count",
               "MSI Score",
               "Fraction Genome Altered",
               "Tumor Purity"),
  Skewness = round(sapply(df[continuous_vars],
                          function(x) skewness(x, na.rm = TRUE)), 4),
  Interpretation = sapply(df[continuous_vars], function(x) {
    s <- skewness(x, na.rm = TRUE)
    if(abs(s) < 0.5) return("Symmetric")
    else if(s > 0.5) return("Right-skewed")
    else return("Left-skewed")
  }),
  Transformation = c("Log transformation applied",
                     "Recategorized to MSI Type Combined",
                     "No transformation needed",
                     "No transformation needed")
)

rownames(skewness_check) <- NULL

# Force table counter to start at 3
knitr::opts_chunk$set(table.number = 3)

kable(skewness_check,
```

```r
      col.names = c("Variable", "Skewness",
                    "Interpretation", "Action Taken"),
      align = c("l", "c", "c", "l"),
      caption = "Skewness Assessment and Transformation Decisions",
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE,
                latex_options = c("scale_down", "hold_position")) %>%
  row_spec(c(1,2), bold = TRUE, background = "#fff3e0") %>%
  footnote(general = "MSI Score recategorized into MSI Type Combined due to severe right
           threeparttable = TRUE)

# Apply log transformation
df$mutation_count_log <- log(df$mutation_count + 1)

# Combine rare MSI Type categories for modeling
df$msi_type_combined <- dplyr::case_when(
  df$msi_type == "Stable" ~ "Stable",
  df$msi_type == "Instable" ~ "Instable",
  TRUE ~ "Other"
)

df$msi_type_combined <- factor(df$msi_type_combined,
                               levels = c("Stable", "Instable", "Other"))
# Plot : Distribution of continuous predictors (before transformation)

par(mfrow=c(2,2))

hist(df$mutation_count, breaks=30, col="steelblue", border="white",
     main="Mutation Count", xlab="Mutation Count", las=1)

hist(df$fraction_genome_altered, breaks=30, col="coral", border="white",
     main="Fraction Genome Altered", xlab="FGA", las=1)

hist(df$msi_score, breaks=30, col="seagreen", border="white",
     main="MSI Score", xlab="MSI Score", las=1)


hist(df$tumor_purity, breaks=30, col="orange", border="white",
     main="Tumor Purity", xlab="Purity (%)", las=1)

# Add figure caption at the bottom
mtext("Figure 1: Distribution of continuous predictors before transformation.",
      side=1, line=1, cex=0.9, adj=0.5, font=2,
      outer=TRUE)   # ← outer=TRUE puts it outside all plots

# Plot : Log-transformed distributions
```

```r
hist(df$mutation_count_log, breaks=30, col="steelblue", border="white",
     main="Figure 2.Log-Transformed Mutation Count",
     xlab="log(Mutation Count + 1)",
     ylab="Frequency",
     las=1,
     xlim=c(0, 4.5),
     ylim=c(0, 1000),
     xaxs="i",
     yaxs="i",
     xaxt="n")  # ← suppress default x axis

# Add custom x axis with more precise breaks
axis(1, at=seq(0, 4.5, by=0.5))  # ← marks at 0, 0.5, 1.0, 1.5...

box(bty="l")

#  Create Analysis-Ready Datasets

# Dataset : For Research Question (Survival prediction)

df_survival <- df %>%
  select(survival_binary,
         mutation_count, mutation_count_log,
         fraction_genome_altered,
         msi_type_combined,
         sample_type, is_metastatic,
         cancer_type_simple,
         tumor_purity) %>%
  filter(complete.cases(.))

# Complete dataset with all variables

df_complete <- df %>%
  filter(!is.na(survival_binary),
         !is.na(mutation_count_log),
         !is.na(fraction_genome_altered),
         !is.na(tumor_purity),
         !is.na(msi_type_combined))

var_summary <- data.frame(
  Variable = c("Overall Survival Status",
               "Log Mutation Count",
               "Fraction Genome Altered",
               "Tumor Purity",
               "MSI Type Combined",
               "Metastatic Status",
               "Cancer Type"),
```

```r
  Type = c("Binary",
           "Continuous",
           "Continuous",
           "Continuous",
           "Categorical",
           "Binary",
           "Categorical"),
  Role = c("Outcome",
           "Primary Predictor",
           "Primary Predictor",
           "Confounder",
           "Primary Predictor",
           "Primary Predictor",
           "Confounder"),
  Details = c("Deceased (1) / Living (0)",
              "log(Mutation Count + 1)",
              "Range: 0 to 1",
              "Range: 10% to 100%",
              "Stable / Instable / Other",
              "Metastatic (1) / Primary (0)",
              "Invasive Ductal / Lobular / Mixed / Other")
)

kable(var_summary,
      col.names = c("Variable", "Type", "Role", "Details"),
      align = c("l", "c", "c", "l"),
      caption = "Summary of Variables Used in Analysis",
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE) %>%
  row_spec(1, bold = TRUE, background = "#e8f4f8") %>%  # highlight outcome
  column_spec(1, bold = TRUE)
# Outcome Variable Distributions
table_survival <- table(df$overall_survival_status)
print(table_survival)
prop_survival <- prop.table(table_survival) * 100

table_sample <- table(df$sample_type)
print(table_sample)
prop_sample <- prop.table(table_sample) * 100

# Predictor Variable Distributions
print(summary(df$mutation_count_log))
print(summary(df$fraction_genome_altered))
print(summary(df$tumor_purity))

# MSI Type Combined distribution
```

```r
table_msi <- table(df$msi_type_combined)
prop_msi <- prop.table(table_msi) * 100
cat("\nMSI Type Distribution:\n")
print(cbind(
  Count = table_msi,
  Percentage = round(prop_msi, 2)
))

par(mfrow=c(2,2), oma=c(3,0,0,0))

boxplot(mutation_count_log ~ survival_binary, data=df_complete,
        col=c("lightgreen", "lightcoral"),
        names=c("Living", "Deceased"),
        main="Log Mutation Count by Survival Status",
        ylab="Mutation Count", las=1)

boxplot(fraction_genome_altered ~ survival_binary, data=df_complete,
        col=c("lightgreen", "lightcoral"),
        names=c("Living", "Deceased"),
        main="FGA by Survival Status",
        ylab="Fraction Genome Altered", las=1)

boxplot(tumor_purity ~ survival_binary, data=df_complete,
        col=c("lightgreen", "lightcoral"),
        names=c("Living", "Deceased"),
        main="Tumor Purity by Survival Status",
        ylab="Tumor Purity (%)", las=1)

# barplot for MSI Type
msi_surv_table <- table(df_complete$msi_type_combined,
                        df_complete$survival_binary)

bp <- barplot(prop.table(msi_surv_table, margin=1) * 100,
        beside=TRUE,
        col=c("lightgreen", "lightcoral", "lightyellow"),
        names.arg=c("Living", "Deceased"),
        main="MSI Type by Survival Status",
        ylab="Percentage (%)",
        las=1,
        legend=FALSE,
        xlim=c(0, 12))

legend(x=10, y=70,
       legend=c("Stable", "Instable", "Other"),
       fill=c("lightgreen", "lightcoral", "lightyellow"),
       cex=0.7,
       xpd=TRUE)
```

```r
mtext("Figure 3: Distribution of predictors by survival status (Living vs Deceased).",
      side=1, line=1, cex=0.9, adj=0.5, font=2, outer=TRUE)
# Correlation Analysis
cor_vars <- df_complete %>%
  select(mutation_count_log,
         fraction_genome_altered,
         tumor_purity) %>%
  na.omit()

cor_matrix <- cor(cor_vars)

# Convert to table format
cor_table <- as.data.frame(round(cor_matrix, 3))
rownames(cor_table) <- c("Log Mutation Count",
                         "Fraction Genome Altered",
                         "Tumor Purity")
colnames(cor_table) <- c("Log Mutation Count",
                         "Fraction Genome Altered",
                         "Tumor Purity")

kable(cor_table,
      caption = "Correlation Matrix of Continuous Predictors",
      align = "c") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE) %>%
  footnote(general = "All correlations |r| < 0.5. No multicollinearity concerns.")

# Check for high correlations - result shown in footnote above
high_cor <- which(abs(cor_matrix) > 0.7 & cor_matrix != 1, arr.ind = TRUE)
if(nrow(high_cor) > 0) {
  for(i in 1:nrow(high_cor)) {
    var1 <- rownames(cor_matrix)[high_cor[i,1]]
    var2 <- colnames(cor_matrix)[high_cor[i,2]]
    r_val <- cor_matrix[high_cor[i,1], high_cor[i,2]]
    cat(sprintf("High correlation: %s ~ %s = %.3f\n", var1, var2, r_val))
  }
}


# Plot 4: Correlation matrix
library(corrplot)
corrplot(cor_matrix, method="color", type="upper",
         tl.col="black", tl.srt=45,
         addCoef.col = "black", number.cex=0.7,
         col=colorRampPalette(c("blue", "white", "red"))(200),
         title="Fig 4 Correlation Matrix of Continuous Predictors",
         mar=c(0,0,2,0))
```

```r
# Bivariate Relationships

# Mutation count by survival status

tapply(df_complete$mutation_count_log, df_complete$survival_binary, summary)

# T-test
t_test_mut <- t.test(mutation_count_log ~ survival_binary, data = df_complete)

t_test_mut

# Fraction Genome Altered by survival status

tapply(df_complete$fraction_genome_altered, df_complete$survival_binary, summary)

t_test_fga <- t.test(fraction_genome_altered ~ survival_binary, data = df_complete)

# print FGA t-test
t_test_fga
# Categorical Associations

# Cancer type by survival
table_cancer_surv <- table(df_complete$cancer_type_simple, df_complete$survival_binary)
print(table_cancer_surv)
print(prop.table(table_cancer_surv, margin=1) * 100)

fisher_cancer <- fisher.test(table_cancer_surv)
fisher_cancer

chisq_cancer <- chisq.test(table_cancer_surv)
chisq_cancer


# Sample type by survival

table_sample_surv <- table(df_complete$sample_type, df_complete$survival_binary)
print(table_sample_surv)
print(prop.table(table_sample_surv, margin=1) * 100)

chisq_sample <- chisq.test(table_sample_surv[c("Primary", "Metastasis"),])
chisq_sample
# Plot 5: Survival by Sample Type
library(ggplot2)
sample_surv_df <- as.data.frame(table_sample_surv)
colnames(sample_surv_df) <- c("Sample_Type", "Survival", "Count")

ggplot(sample_surv_df, aes(x=Sample_Type, y=Count, fill=factor(Survival))) +
```

```r
  geom_bar(stat="identity", position="dodge") +
  scale_fill_manual(values=c("0"="lightgreen", "1"="lightcoral"),
                    labels=c("Living", "Deceased"),
                    name="Survival Status") +
  labs(title="Fig 5 Survival Status by Sample Type",
       x="Sample Type", y="Count") +
  theme_minimal() +
  theme(text=element_text(size=12))

par(mfrow=c(1,2))

# Log Mutation count vs FGA
plot(df_complete$mutation_count_log, df_complete$fraction_genome_altered,
     pch=16, col=rgb(0,0,1,0.3),
     xlab="log(Mutation Count + 1)", ylab="Fraction Genome Altered",
     main="Log Mutation Count vs FGA")
abline(lm(fraction_genome_altered ~ mutation_count_log, data=df_complete),
       col="red", lwd=2)

# Tumor purity vs Log Mutation count
plot(df_complete$tumor_purity, df_complete$mutation_count_log,
     pch=16, col=rgb(0.5,0,0.5,0.3),
     xlab="Tumor Purity (%)", ylab="log(Mutation Count + 1)",
     main="Tumor Purity vs Log MC")  # ← shortened title
abline(lm(mutation_count_log ~ tumor_purity, data=df_complete),
       col="darkgreen", lwd=2)
model1_adjusted <- glm(survival_binary ~
                         mutation_count_log +
                         fraction_genome_altered +
                         msi_type_combined +
                         is_metastatic +
                         cancer_type_simple +
                         tumor_purity,
                       data = df_survival,
                       family = binomial(link = "logit"))
summary(model1_adjusted)
# Calculate odds ratios and 95% CI
or_adj <- exp(coef(model1_adjusted))
or_ci_adj <- exp(confint(model1_adjusted))

results_adj <- data.frame(
  Predictor = names(coef(model1_adjusted)),
  Coefficient = coef(model1_adjusted),
  OR = or_adj,
  OR_Lower = or_ci_adj[,1],
  OR_Upper = or_ci_adj[,2],
  P_value = summary(model1_adjusted)$coefficients[,4]
```

```r
)

# Extract key genomic predictors
genomic_preds <- results_adj[grepl("mutation|fraction|msi_type",
                                   results_adj$Predictor,
                                   ignore.case=TRUE), ]

for(i in 1:nrow(genomic_preds)) {
  pred <- genomic_preds$Predictor[i]
  or_val <- genomic_preds$OR[i]
  ci_lower <- genomic_preds$OR_Lower[i]
  ci_upper <- genomic_preds$OR_Upper[i]
  p_val <- genomic_preds$P_value[i]

  cat(sprintf("\n%d. %s\n", i, pred))
  cat(sprintf("   OR = %.3f (95%% CI: %.3f - %.3f), p = %s\n",
              or_val, ci_lower, ci_upper, format.pval(p_val, digits=3)))

  if(p_val < 0.05) {
    if(grepl("mutation", pred)) {
      cat("    SIGNIFICANT: Higher mutation burden predicts increased mortality\n")
      cat("   → After adjusting for metastatic status, cancer type, and tumor purity,\n"
      cat("     each unit increase in log(mutation count) increases odds of death by",
          round((or_val-1)*100, 1), "%\n")
    } else if(grepl("fraction", pred)) {
      cat("    SIGNIFICANT: Greater genomic instability predicts worse survival\n")
      cat("   → Each 0.1 increase in FGA increases odds of death by",
          round((or_val^0.1-1)*100, 1), "%\n")
    } else if(grepl("msi_type", pred)) {
      cat("    SIGNIFICANT: MSI type category is independently predictive\n")
    }
  } else {
    cat("    NOT SIGNIFICANT after adjusting for confounders\n")
  }
}
# Multicollinearity (VIF)

vif_values <- vif(model1_adjusted)
vif_values

# Goodness of Fit (Hosmer-Lemeshow Test)

hl_test <- hoslem.test(model1_adjusted$y, fitted(model1_adjusted), g=10)
hl_test


model2_interaction <- glm(survival_binary ~
```

```r
                              mutation_count_log * is_metastatic +  # interaction
                              fraction_genome_altered +
                              msi_type_combined +
                              cancer_type_simple +
                              tumor_purity,
                            data = df_survival,
                            family = binomial(link = "logit"))
summary(model2_interaction)
# AIC comparison
AIC(model1_adjusted, model2_interaction)

# Likelihood ratio test
anova(model1_adjusted, model2_interaction, test = "Chisq")
hl_test2 <- hoslem.test(model2_interaction$y,
                        fitted(model2_interaction), g = 10)
hl_test2
model3_interaction <- glm(survival_binary ~
                              mutation_count_log +
                              fraction_genome_altered * is_metastatic +  # new interactio
                              msi_type_combined +
                              cancer_type_simple +
                              tumor_purity,
                            data = df_survival,
                            family = binomial(link = "logit"))
summary(model3_interaction)

# Compare to model1

aic_table_clean <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3"),
  Description = c(
    "Main effects only",
    "Main effects + MC x Meta",
    "Main effects + FGA x Meta"),
  AIC = c(round(AIC(model1_adjusted), 2),
          round(AIC(model2_interaction), 2),
          round(AIC(model3_interaction), 2)),
  Delta_AIC = c(0.00, 1.74, -4.37),
  HL_pvalue = c("0.006", "0.010", "0.269"),
  Selected = c("", "", "Yes")
)

kable(aic_table_clean,
      col.names = c("Model", "Description",
                    "AIC", "Delta AIC", "HL p-value", "Selected"),
      align = c("l","l","c","c","c","c"),
      caption = "Model Comparison and Selection",
```

```r
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE) %>%
  row_spec(3, bold = TRUE, background = "#e8f5e9")
# Print which model is best
cat(sprintf("\n  Best model: Model 3 (lowest AIC = %.3f)\n",
            AIC(model3_interaction)))
cat(sprintf("  Model 3 improves AIC by %.3f over Model 1\n",
            AIC(model1_adjusted) - AIC(model3_interaction)))

#LRT confirming model3 is better than model1

anova(model1_adjusted, model3_interaction, test = "Chisq")

# HL test
hl_test3 <- hoslem.test(model3_interaction$y,
                        fitted(model3_interaction), g = 10)
hl_test3
# Pseudo R-squared
null_dev <- model3_interaction$null.deviance
resid_dev <- model3_interaction$deviance
mcfadden_r2 <- 1 - (resid_dev / null_dev)

# Cox-Snell R-squared
n <- nrow(df_survival)
cox_snell_r2 <- 1 - exp((resid_dev - null_dev) / n)

# Nagelkerke R-squared
nagelkerke_r2 <- cox_snell_r2 / (1 - exp(-null_dev / n))

# Clean table
r2_table <- data.frame(
  Measure = c("McFadden's R²",
              "Cox-Snell R²",
              "Nagelkerke R²"),
  Value = c(round(mcfadden_r2, 4),
            round(cox_snell_r2, 4),
            round(nagelkerke_r2, 4)),
  Interpretation = c(
    "Acceptable if > 0.2",
    "Conservative lower bound",
    "Adjusted, comparable to linear R²")
)

kable(r2_table,
      col.names = c("Measure", "Value", "Interpretation"),
      align = c("l", "c", "l"),
```

```r
      caption = "Pseudo R-squared Statistics - Final Model") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE)
or_adj3 <- exp(coef(model3_interaction))
or_ci_adj3 <- exp(confint(model3_interaction))
results_adj3 <- data.frame(
  Predictor = c("Intercept",
                "Log Mutation Count",
                "Fraction Genome Altered",
                "Metastatic Status",
                "MSI Type: Instable",
                "MSI Type: Other",
                "Cancer Type: Invasive Lobular",
                "Cancer Type: Mixed",
                "Cancer Type: Other",
                "Tumor Purity",
                "FGA x Metastatic"),
  Coefficient = round(coef(model3_interaction), 4),
  OR = round(or_adj3, 4),
  OR_Lower = round(or_ci_adj3[,1], 4),
  OR_Upper = round(or_ci_adj3[,2], 4),
  P_value = round(summary(model3_interaction)$coefficients[,4], 6)
)

rownames(results_adj3) <- NULL

kable(results_adj3,
      col.names = c("Predictor", "Coefficient",
                    "OR", "95% CI Lower",
                    "95% CI Upper", "P-value"),
      align = c("l","c","c","c","c","c"),
      caption = "Final Model Coefficients and Odds Ratios",
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped","hover","bordered"),
                full_width = FALSE) %>%
  row_spec(which(results_adj3$P_value < 0.05),
           bold = TRUE, background = "#e8f5e9")
# Interaction Plot - FGA x Metastatic Status
fga_seq <- seq(min(df_survival$fraction_genome_altered),
               max(df_survival$fraction_genome_altered),
               length.out = 100)

newdata_primary <- data.frame(
  fraction_genome_altered = fga_seq,
  is_metastatic = 0,
  mutation_count_log = mean(df_survival$mutation_count_log, na.rm=TRUE),
  msi_type_combined = factor("Stable",
```

```r
                              levels = c("Stable","Instable","Other")),
  cancer_type_simple = "Invasive Ductal",
  tumor_purity = mean(df_survival$tumor_purity, na.rm=TRUE)
)

newdata_meta <- data.frame(
  fraction_genome_altered = fga_seq,
  is_metastatic = 1,
  mutation_count_log = mean(df_survival$mutation_count_log, na.rm=TRUE),
  msi_type_combined = factor("Stable",
                              levels = c("Stable","Instable","Other")),
  cancer_type_simple = "Invasive Ductal",
  tumor_purity = mean(df_survival$tumor_purity, na.rm=TRUE)
)

pred_primary <- predict(model3_interaction,
                        newdata = newdata_primary,
                        type = "response")
pred_meta <- predict(model3_interaction,
                     newdata = newdata_meta,
                     type = "response")

par(oma=c(3,0,0,0))  # ← add this before plot

plot(fga_seq, pred_primary,
     type = "l", col = "blue", lwd = 2,
     ylim = c(0, 1),
     xlab = "Fraction Genome Altered (FGA)",
     ylab = "Predicted Probability of Death",
     main = "Interaction Effect: FGA x Metastatic Status")
lines(fga_seq, pred_meta, col = "red", lwd = 2)
legend("topleft",
       legend = c("Primary Tumor", "Metastatic Tumor"),
       col = c("blue", "red"), lwd = 2)
abline(h = 0.5, lty = 2, col = "gray")
grid()

mtext("Figure 7: Interaction effect of FGA and metastatic status",
      side=1, line=1, cex=0.9, adj=0.5, font=2, outer=TRUE)
vif(model3_interaction)
# Influential Observations

# Cook's Distance
cooks_d3 <- cooks.distance(model3_interaction)
influential_threshold3 <- 4 / nrow(df_survival)

n_influential3 <- sum(cooks_d3 > influential_threshold3)
```

```r
percentage3 <- round(100 * n_influential3 / nrow(df_survival), 2)

plot(cooks_d3, type="h",
     main="Cook's Distance - Final Model",
     ylab="Cook's Distance", xlab="Observation Index")
abline(h=influential_threshold3, col="red", lty=2)

par(mfrow=c(1,2))

# Model Discrimination (ROC Curve)

# Calculate predicted probabilities - Final Model
pred_prob3 <- predict(model3_interaction, type = "response")

# ROC curve
roc_obj3 <- roc(df_survival$survival_binary, pred_prob3)
auc_value3 <- auc(roc_obj3)


if(auc_value3 > 0.8) {
  cat("  EXCELLENT discrimination (AUC > 0.8)\n\n")
} else if(auc_value3 > 0.7) {
  cat("  ACCEPTABLE discrimination (AUC > 0.7)\n\n")
} else if(auc_value3 > 0.6) {
  cat("  POOR discrimination (AUC 0.6-0.7)\n")
  cat("SOLUTION: Consider adding more predictors or interaction terms\n\n")
} else {
  cat("  VERY POOR discrimination (AUC < 0.6)\n")
  cat("SOLUTION: Model may need major revision\n\n")
}

# Plot ROC curve
plot(roc_obj3, main="ROC Curve - Final Model (FGA × Metastatic)",
     col="blue", lwd=2, print.auc=TRUE, print.auc.y=0.4)
abline(a=0, b=1, lty=2, col="gray")


# Plot Calibration Plot - Final Model

cal_data3 <- data.frame(
  predicted = pred_prob3,
  observed = df_survival$survival_binary
)

cal_data3$decile <- cut(cal_data3$predicted,
                        breaks=quantile(cal_data3$predicted,
                                        probs=seq(0,1,0.1)),
```

```r
                        include.lowest=TRUE, labels=1:10)

cal_summary3 <- cal_data3 %>%
  group_by(decile) %>%
  summarise(mean_predicted = mean(predicted),
            mean_observed = mean(observed),
            n = n())

plot(cal_summary3$mean_predicted, cal_summary3$mean_observed,
     xlim=c(0,1), ylim=c(0,1), pch=19, col="blue", cex=1.5,
     xlab="Predicted Probability", ylab="Observed Proportion",
     main="Calibration Plot - Final Model")
abline(a=0, b=1, lty=2, col="red", lwd=2)
grid()


par(mfrow=c(1,2), mar=c(4,4,3,1), oma=c(3,0,0,0))  # ← add oma

# ROC Curve - zoomed in
plot(roc_obj3,
     main="ROC Curve - Final Model",
     col="blue", lwd=2,
     print.auc=TRUE,
     print.auc.y=0.4,
     print.auc.cex=0.8,
     cex.main=0.9,
     cex.axis=0.8,
     cex.lab=0.8)
abline(a=0, b=1, lty=2, col="gray")

# Calibration Plot
cal_data3 <- data.frame(
  predicted = pred_prob3,
  observed = df_survival$survival_binary
)
cal_data3$decile <- cut(cal_data3$predicted,
                        breaks=quantile(cal_data3$predicted,
                                        probs=seq(0,1,0.1)),
                        include.lowest=TRUE, labels=1:10)
cal_summary3 <- cal_data3 %>%
  group_by(decile) %>%
  summarise(mean_predicted = mean(predicted),
            mean_observed = mean(observed),
            n = n())
plot(cal_summary3$mean_predicted, cal_summary3$mean_observed,
     xlim=c(0,1), ylim=c(0,1), pch=19, col="blue", cex=1.2,
     xlab="Predicted Probability", ylab="Observed Proportion",
```

```r
     main="Calibration Plot - Final Model",
     cex.main=0.9,
     cex.axis=0.8,
     cex.lab=0.8)
abline(a=0, b=1, lty=2, col="red", lwd=2)
grid()

# Add figure caption at bottom
mtext("Figure 8: ROC curve (AUC = 0.741) and calibration plot for the final model.",
      side=1, line=1, cex=0.9, adj=0.5, font=2, outer=TRUE)

cal_data3 <- data.frame(
  predicted = pred_prob3,
  observed = df_survival$survival_binary
)

cal_data3$decile <- cut(cal_data3$predicted,
                        breaks=quantile(cal_data3$predicted,
                                        probs=seq(0,1,0.1)),
                        include.lowest=TRUE, labels=1:10)

cal_summary3 <- cal_data3 %>%
  group_by(decile) %>%
  summarise(mean_predicted = mean(predicted),
            mean_observed = mean(observed),
            n = n())

plot(cal_summary3$mean_predicted, cal_summary3$mean_observed,
     xlim=c(0,1), ylim=c(0,1), pch=19, col="blue", cex=1.5,
     xlab="Predicted Probability", ylab="Observed Proportion",
     main="Calibration Plot - Final Model")
abline(a=0, b=1, lty=2, col="red", lwd=2)
grid()


# 6. Residual Diagnostics

dev_resid3 <- residuals(model3_interaction, type="deviance")
par(mfrow=c(2,2), oma=c(3,0,0,0))  # ← add oma for bottom space

plot(fitted(model3_interaction), dev_resid3,
     main="Residuals vs Fitted",
     xlab="Fitted Values", ylab="Deviance Residuals",
     pch=20, col=rgb(0,0,1,0.3))
abline(h=0, col="red", lty=2)

qqnorm(dev_resid3, main="Q-Q Plot", pch=20, col=rgb(0,0,1,0.3))
```

```r
qqline(dev_resid3, col="red", lwd=2)

plot(fitted(model3_interaction), sqrt(abs(dev_resid3)),
     main="Scale-Location",
     xlab="Fitted Values", ylab="√|Deviance Residuals|",
     pch=20, col=rgb(0,0,1,0.3))

plot(cooks_d3, type="h",
     main="Cook's Distance",
     ylab="Cook's Distance", xlab="Observation Index")
abline(h=influential_threshold3, col="red", lty=2)

# Add figure caption at bottom
mtext("Figure 9: Model diagnostic plots for the final model (FGA x Metastatic).",
      side=1, line=1, cex=0.9, adj=0.5, font=2, outer=TRUE)

# Plot : Distribution of continuous predictors (before transformation)

par(mfrow=c(2,2))

hist(df$mutation_count, breaks=30, col="steelblue", border="white",
     main="Mutation Count", xlab="Mutation Count", las=1)

hist(df$fraction_genome_altered, breaks=30, col="coral", border="white",
     main="Fraction Genome Altered", xlab="FGA", las=1)

hist(df$msi_score, breaks=30, col="seagreen", border="white",
     main="MSI Score", xlab="MSI Score", las=1)


hist(df$tumor_purity, breaks=30, col="orange", border="white",
     main="Tumor Purity", xlab="Purity (%)", las=1)

# Add figure caption at the bottom
mtext("Figure 1: Distribution of continuous predictors before transformation.",
      side=1, line=1, cex=0.9, adj=0.5, font=2,
      outer=TRUE)   # ← outer=TRUE puts it outside all plots

# Correlation matrix

library(corrplot)
corrplot(cor_matrix, method="color", type="upper",
         tl.col="black", tl.srt=45,
         addCoef.col = "black", number.cex=0.7,
         col=colorRampPalette(c("blue", "white", "red"))(200),
         title="Correlation Matrix of Continuous Predictors",
         mar=c(0,0,2,0))
```

```r
library(ggplot2)
sample_surv_df <- as.data.frame(table_sample_surv)
colnames(sample_surv_df) <- c("Sample_Type", "Survival", "Count")

ggplot(sample_surv_df, aes(x=Sample_Type, y=Count, fill=factor(Survival))) +
  geom_bar(stat="identity", position="dodge") +
  scale_fill_manual(values=c("0"="lightgreen", "1"="lightcoral"),
                    labels=c("Living", "Deceased"),
                    name="Survival Status") +
  labs(title="Survival Status by Sample Type",
       x="Sample Type", y="Count") +
  theme_minimal() +
  theme(text=element_text(size=12))

par(mfrow=c(1,2), oma=c(3,0,0,0))
# Log Mutation count vs FGA
plot(df_complete$mutation_count_log, df_complete$fraction_genome_altered,
     pch=16, col=rgb(0,0,1,0.3),
     xlab="log(Mutation Count + 1)", ylab="Fraction Genome Altered",
     main="Log Mutation Count vs FGA")
abline(lm(fraction_genome_altered ~ mutation_count_log, data=df_complete),
       col="red", lwd=2)

# Tumor purity vs Log Mutation count
plot(df_complete$tumor_purity, df_complete$mutation_count_log,
     pch=16, col=rgb(0.5,0,0.5,0.3),
     xlab="Tumor Purity (%)", ylab="log(Mutation Count + 1)",
     main="Tumor Purity vs Log MC")  # ← shortened title
abline(lm(mutation_count_log ~ tumor_purity, data=df_complete),
       col="darkgreen", lwd=2)
# Add figure caption at bottom
mtext("Figure 6: Scatter plots of continuous predictors showing weak linear relationship
      side=1, line=1, cex=0.9, adj=0.5, font=2, outer=TRUE)
# Missing data assessment
missing_summary <- data.frame(
  Variable = names(df),
  Missing_Count = sapply(df, function(x) sum(is.na(x))),
  Missing_Percent = round(sapply(df, function(x) 100 * sum(is.na(x)) / length(x)), 2)
) %>%
  arrange(desc(Missing_Count))

rownames(missing_summary) <- NULL

kable(missing_summary,
      col.names = c("Variable", "Missing Count", "Missing (%)"),
      align = c("l", "c", "c"),
      caption = "Missing Data Assessment",
```

```r
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE,
                latex_options = "hold_position") %>%
  row_spec(which(missing_summary$Missing_Percent > 0),
           background = "#fff3e0") %>%
  footnote(general = "All variables have less than 3% missing data. Listwise deletion ap
           threeparttable = TRUE)

# Variables with >10% missing
high_missing <- missing_summary$Variable[missing_summary$Missing_Percent > 10]
# Function to detect outliers using IQR method
detect_outliers <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower <- Q1 - 3 * IQR
  upper <- Q3 + 3 * IQR
  outliers <- x < lower | x > upper
  return(sum(outliers, na.rm = TRUE))
}

continuous_vars <- c("mutation_count",
                     "msi_score",
                     "fraction_genome_altered",
                     "tumor_purity")

outlier_counts <- data.frame(
  Variable = c("Mutation Count",
               "MSI Score",
               "Fraction Genome Altered",
               "Tumor Purity"),
  N_Outliers = sapply(df[continuous_vars], detect_outliers),
  Percent = round(100 * sapply(df[continuous_vars],
                               detect_outliers) / nrow(df), 2)
)

rownames(outlier_counts) <- NULL

kable(outlier_counts,
      col.names = c("Variable", "N Outliers", "Percentage (%)"),
      align = c("l", "c", "c"),
      caption = "Outlier Detection Using 3xIQR Method",  # ← changed × to x
      row.names = FALSE) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE) %>%
  row_spec(which(outlier_counts$Percent > 0),
```

```r
            bold = TRUE, background = "#fff3e0")
# Pseudo R-squared
null_dev <- model3_interaction$null.deviance
resid_dev <- model3_interaction$deviance
mcfadden_r2 <- 1 - (resid_dev / null_dev)

# Cox-Snell R-squared
n <- nrow(df_survival)
cox_snell_r2 <- 1 - exp((resid_dev - null_dev) / n)

# Nagelkerke R-squared
nagelkerke_r2 <- cox_snell_r2 / (1 - exp(-null_dev / n))

# Clean table
r2_table <- data.frame(
  Measure = c("McFadden's R²",
              "Cox-Snell R²",
              "Nagelkerke R²"),
  Value = c(round(mcfadden_r2, 4),
            round(cox_snell_r2, 4),
            round(nagelkerke_r2, 4)),
  Interpretation = c(
    "Acceptable if > 0.2",
    "Conservative lower bound",
    "Adjusted, comparable to linear R²")
)

kable(r2_table,
      col.names = c("Measure", "Value", "Interpretation"),
      align = c("l", "c", "l"),
      caption = "Pseudo R-squared Statistics - Final Model") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE)
```